

Setup Guide for the IMERG GPM Notebook

Setting up the environment, launching Jupyter Lab, and configuring your NASA Earthdata account

This guide explains, step by step, how to prepare your computer to use the **IMERG_notebook.ipynb** notebook: opening the command prompt, creating the dedicated Python environment from the **environment.yml** file, launching Jupyter Lab, and creating the free NASA Earthdata account needed to download the satellite data.

Follow the sections in order. Commands to type are shown in the grey boxes.

Note: This guide has been tested step by step on Windows. The macOS and Linux instructions follow the same logic and use the same commands as their official documentation, but were generated with AI assistance and have not been independently tested — if something doesn't behave as described, the Windows steps show the intended logic.

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1. Requirements and necessary files

Before starting, make sure you have the following files and folders on your computer, saved in the same folder (for example a folder called **imerg_course** on your Desktop):

- **IMERG_notebook.ipynb** — the Jupyter notebook with all the exercises
- **environment.yml** — the file describing the Python environment with all required libraries
- a **shapefile/** subfolder containing the basin shapefile (`.shp` plus its `.shx`, `.dbf`, `.prj`, ... companion files) — needed from Section 5 of the notebook onwards
- this PDF guide

You also need an internet connection, Anaconda already installed on your computer, and a free NASA Earthdata account with a `.netrc` file configured in this same folder (explained in sections 4 and 5). This should be done before opening the notebook.

Note: Don't have Anaconda yet? Download the installer for your operating system from anaconda.com/download and run it with the default options. On macOS/Linux you can also install it straight from the terminal: download the `.sh` installer for your OS from repo.anaconda.com/archive/, then run `bash <downloaded-file>.sh` and follow the on-screen prompts.

Warning: The example **La_Bruna** catchment shapefile is located in **central Italy (Umbria region, Valnerina area)**, roughly between 12.58–12.69 °E and 42.76–42.82 °N. When you draw the area of interest on the interactive map in Section 1 of the notebook, make sure your rectangle **includes this area** — otherwise the basin will fall outside the downloaded IMERG data, and the basin-averaging steps (from Section 5 onwards) will fail or return empty results.

2. Open the command prompt

Windows

- 1 Click the Start menu
- 2 Type *Anaconda Prompt*
- 3 Click on the Anaconda Prompt app to open it

macOS

- 1 Open Launchpad, or press Cmd + Space
- 2 Type *Terminal* and press Enter

Linux

- 1 Open the Terminal application from the applications menu, or use the shortcut Ctrl + Alt + T

Note: In all three cases a dark window with a blinking cursor will open: this is where the commands in the next sections need to be typed.

3. Create the Python environment from environment.yml

A Python environment is a self-contained installation of Python plus a specific set of libraries, kept separate from any other Python installation on your computer. Using a dedicated environment for this course means the exact library versions it needs won't clash with other projects, and can be removed cleanly at the end without affecting anything else.

In the command prompt, move to the folder where the environment.yml file is located, using the `cd` (change directory) command. For example, if the folder is on the Desktop and is called `imerg_course`:

```
cd Desktop/imerg_course
```

Then create the environment with the following command (this only needs to be run once; installing the packages may take a few minutes):

```
conda env create -f environment.yml --force
```

Once finished, the environment will be called **imerg_env** (the name is defined inside the yml file).

Note: The `--force` flag tells conda to recreate the environment from scratch if one named **imerg_env** already exists on your computer (for example, if you took this course before, or ran this command previously). Without it, conda would stop with a "prefix already exists" error instead of updating the environment. If this is the first time you create it, the flag has no effect.

Warning: If the conda command is not recognized, Anaconda may not be installed correctly, or you are not using the Anaconda Prompt (on Windows).

4. Create a NASA Earthdata account

Downloading IMERG data requires a free NASA Earthdata Login account. Registration and application authorization only take a few minutes and must be done before creating the `.netrc` file in the next section.

- 1 Open your browser and go to: `urs.earthdata.nasa.gov/users/new`
- 2 Fill in the registration form (username, password, email, country)
- 3 Confirm your email address using the link you receive by email
- 4 Write down your username and password: you will need them to create the `.netrc` file in section 5

Note: The NASA Earthdata account is free and unique: once created, it can be reused for all future notebooks and exercises in the course.

Authorize the NASA GES DISC application

Creating an Earthdata profile is not enough to download IMERG data: you also need to authorize the application that manages the GES DISC archive, otherwise the download will fail with an error.

- 1 Log in to your profile at `urs.earthdata.nasa.gov/profile`
- 2 In the top bar, below your name, open the **Applications** drop-down menu and select **Authorized Apps**
- 3 The **Approved Applications** page will open, showing the list of applications you have already authorized
- 4 Check whether **NASA GESDISC DATA ARCHIVE** is already in the list. If it is, no further action is needed
- 5 If it is not, scroll to the bottom of the page and click the light-blue **APPROVE MORE APPLICATIONS** button
- 6 Search for **NASA GESDISC DATA ARCHIVE**, click **Authorize**, and accept the End User License Agreement (EULA) if prompted

5. Create the `.netrc` file

`.netrc` is a small, standard configuration file (used by many command-line and scientific tools, such as `curl` and the NASA data libraries) that stores login credentials for a remote server, so a program can authenticate automatically instead of asking you to type a username and password every time.

The notebook authenticates with NASA Earthdata by reading credentials from a configuration file called `.netrc`. On all operating systems, including Windows, this file must be located in the same folder as the notebook (the same folder as `IMERG_notebook.ipynb` and `environment.yml`) — not in your user/home folder. Without this file, the login step in the notebook will fail and the data cannot be downloaded. It only needs to be created once, before launching Jupyter Lab.

The simplest and most reliable way to create it is from the command prompt, using the commands below (replace `YOUR_USERNAME` and `YOUR_PASSWORD` with the credentials created in section 4).

Windows (Anaconda Prompt)

Type `cd` followed by the path to your own notebook folder (the same one you used in section 3), then press Enter to move into it. The example below uses `Desktop\imerg_course`: replace it with your actual path if your folder is somewhere else or has a different name.

```
cd Desktop\imerg_course
(
  echo machine urs.earthdata.gov
  echo login YOUR_USERNAME
  echo password YOUR_PASSWORD
) > .netrc
```

Check that the file was created correctly with:

```
type .netrc
```

You should see the three lines (machine, login, password) printed on screen.

Note: Windows File Explorer may not show a file starting with a dot, or may complain if you try to create one by hand — this is normal and only affects the graphical file browser. The command above, run from the Anaconda Prompt, creates the file correctly regardless.

macOS / Linux (Terminal)

Type `cd` followed by the path to your own notebook folder (the same one you used in section 3), then press Enter to move into it. The example below uses `~/Desktop/imerg_course`: replace it with your actual path if your folder is somewhere else or has a different name. The rest of the command creates the `.netrc` file there and sets the correct file permissions right away:

```
cd ~/Desktop/imerg_course
cat > .netrc << EOF
machine urs.earthdata.gov
login YOUR_USERNAME
password YOUR_PASSWORD
EOF
chmod 600 .netrc
```

The `chmod 600` command is important: it makes the file readable only by you, as required by the `.netrc` protocol (some systems refuse to use the file if permissions are too open). To check that it was created correctly:

```
cat .netrc
```

Warning: The `.netrc` file contains your password in plain text: do not share it, do not upload it to public repositories (e.g. GitHub), and do not copy it to other people's computers. Each student should create their own file with their own NASA Earthdata credentials.

6. Activate the environment and run Jupyter Lab

Jupyter Lab is the browser-based application you'll use to open and run the notebook: it lets you write and execute Python code in "cells" and see the results — text, tables, plots, maps — right below each one.

Every time you want to work on the notebook, you need to activate the environment created in section 3:

```
conda activate imerg_env
```

The command prompt will show `(imerg_env)` at the start of the line, confirming the environment is active.

A kernel is the Python process that actually runs your code when you execute a cell in Jupyter — it determines which Python installation, and which packages, are available to the notebook. Registering the `imerg_env` environment as a kernel makes it show up with the right name in Jupyter's kernel picker. This

only needs to be done once, right after creating the environment (you can skip it the next times you use the notebook):

```
python -m ipykernel install --user --name imerg_env --display-name "imerg_env"
```

Now start Jupyter Lab with:

```
jupyter lab
```

A browser window will open automatically showing the Jupyter Lab interface with the files in the current folder. If the browser does not open by itself, copy the address printed in the terminal (something like `http://localhost:8888/...`) and paste it into your browser's address bar.

7. Open the notebook

- 1 In the left panel of Jupyter Lab, find the file `IMERG_notebook.ipynb`
- 2 Double-click it to open it
- 3 Run the cells in order from top to bottom with `Shift + Enter`

Note: The first time you run a cell, Jupyter may ask you to choose a kernel (the Python interpreter): choose the `imerg_env` kernel registered in section 6.

In the notebook's login cell:

```
import earthaccess  
auth = earthaccess.login(strategy="netrc")
```

If the `.netrc` file was created correctly in section 5, the login happens automatically and `Login OK` is printed, with nothing to type.

8. Using the notebook: summary of steps

- 1 Run the setup cells and the Earthdata login (requires the `.netrc` file, in the same folder as the notebook, as created in section 5)
- 2 Choose the temporal resolution (30 min / daily / monthly), the time period, and the processing options (save figures, whole-area maps, whole-area Excel export)
- 3 Draw the rectangle of the area of interest on the interactive map
- 4 Run the search and (optional) download of the IMERG data
- 5 Load and crop the data to the selected area; the notebook prints the raw file structure and converts precipitation to a depth per timestep (mm)
- 6 Optional: generate the mean/max/min maps over the whole selected area and save them to Excel
- 7 Place the basin shapefile in the `shapefile/` subfolder; the notebook validates it (a single polygon is required) and computes its area
- 8 Compare the IMERG grid with the basin, and generate the zoomed-in maps
- 9 Compute basin-averaged precipitation with three methods: `all_touched=True`, `all_touched=False`, and an area-weighted grid-cell method
- 10 Compare the three methods, both as time series and as cumulative precipitation over time
- 11 Select one method to generate the final time series plot and export it to Excel

Note: All plots produced by the notebook (area maps, zoomed maps, time series) can be saved to file if you want to keep them — this is optional and up to you.

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